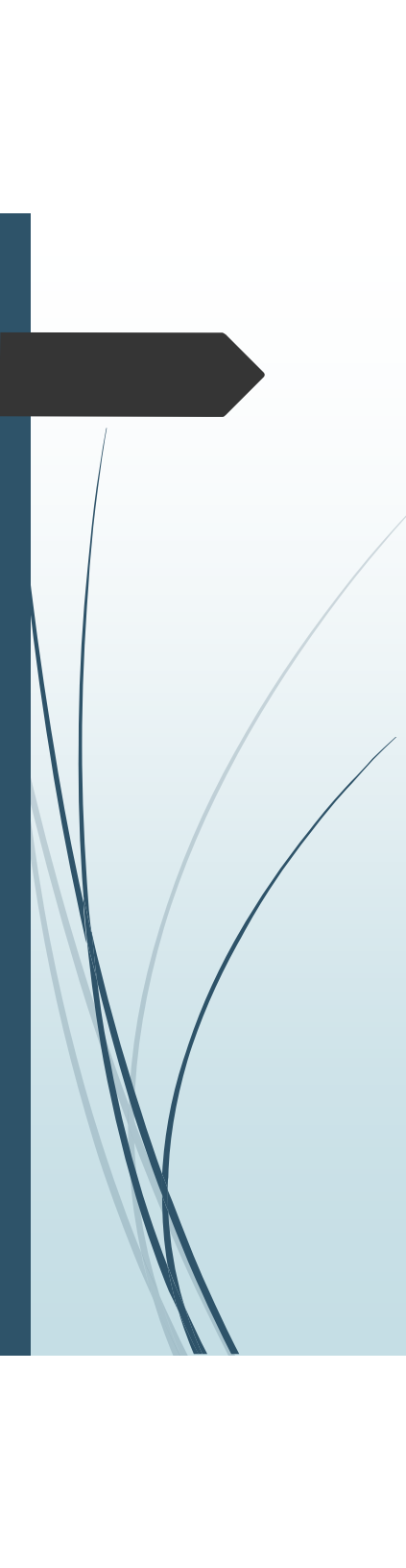




Decision Trees:

Decision Tree Disadvantages:

1. When building decision Tree at its complete depth then then decision tree is learning very well on the train data.
2. On the test data, the decision tree does not generalizes well (poor performance on the new test data).

This results in **overfitting** the model.

- **Overfitting**
- Decision trees can grow **very deep**
- They memorize training data
- High **variance**, low bias



Ensembles of Decision Trees:

A single decision tree overfits. **Ensembles** combine many Decision trees to get more accurate, stable, and generalizable models.

- **How ensembles fix Decision Tree problems**

- ◆ **Reduce variance**

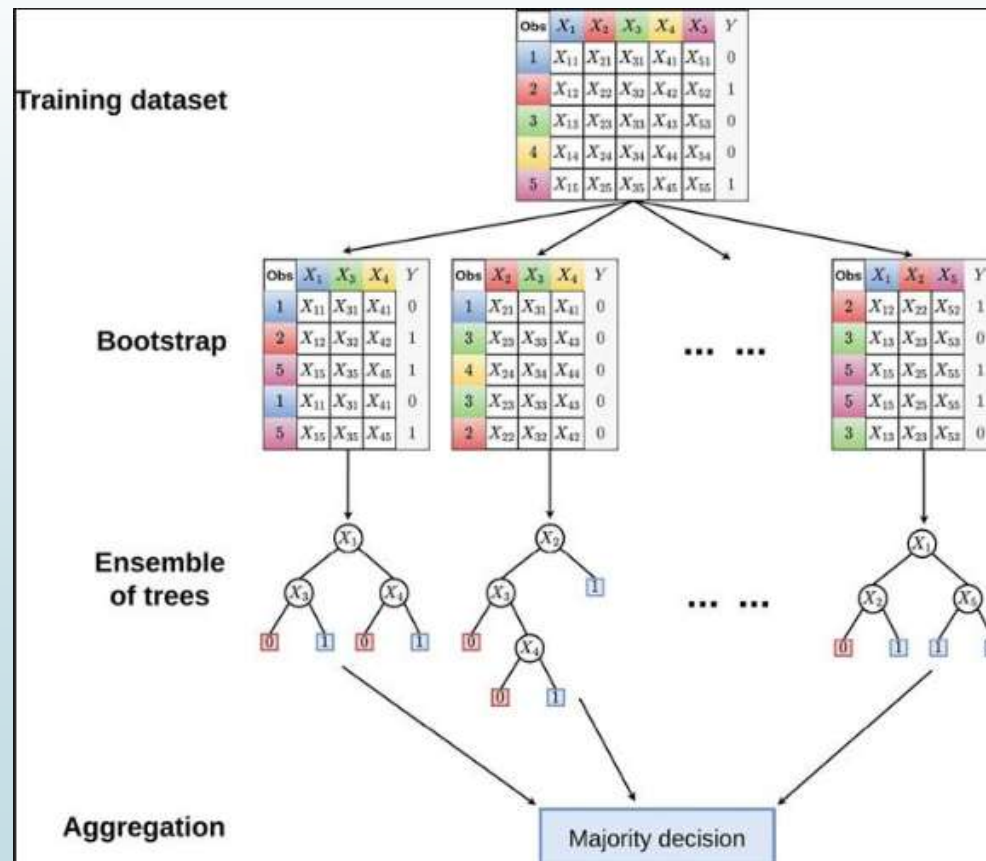
- Each tree makes different mistakes
- Averaging predictions smooths errors

- ◆ **Improve generalization**

- Less sensitive to noise
- Better performance on unseen data

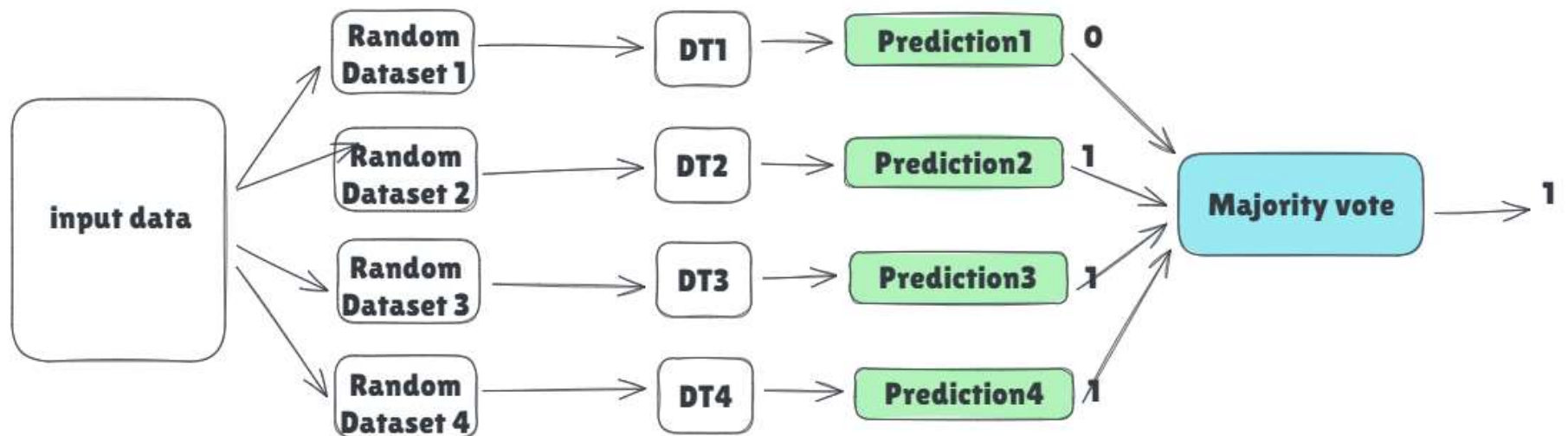
Ensembles of Decision Trees:

1. Random Forest Classifier : Bagging or Bootstrap Samples



Ensembles of Decision Trees:

1. Random Forest Classifier:



Ensembles of Decision Trees:

1. Random Forest Classifier:

Probability-based voting (Actual behavior)

Each tree outputs **class probabilities**, not just labels.

Tree	P(class=0)	P(class=1)
Tree 1	0.20	0.80
Tree 2	0.40	0.60
Tree 3	0.70	0.30
Tree 4	0.55	0.45

Average probabilities

$$P(0) = (0.20 + 0.40 + 0.70 + 0.55) / 4 = 0.4625$$

$$P(1) = (0.80 + 0.60 + 0.30 + 0.45) / 4 = 0.5375$$

Final prediction = class 1